# Design and Analysis of Hybrid Retrieval Systems in NLP

**Group 10**

## Abstract

This project addresses the challenge of misinformation via a hybrid fact-checking approach that combines multistage evidence retrieval and claim classification. Our evidence retrieval pipeline sequentially filters out candidates for evidence using a fine-tuned SBERT model and a Cross-Encoder to balance efficiency and accuracy.

For claim verification, we trained a BERT classifier with multiple evidence concatenation and got a best accuracy of 0.4870 on the development set. Comparative experiments against traditional models like MLP and TextCNN show that BERT is superior. We further experimented with evidence aggregation methods and found that concatenating several evidence passages provides the most robust performance.

Our system was ranked Top-15, demonstrating the efficiency of our approach towards automatic fact-checking.

## 1 Introduction

Misinformation poses a formidable challenge, particularly in domains such as climate science where accuracy of facts is paramount. Automated fact-checking demands effective evidence retrieval and correct claim verification but is challenging because of noisy evidence and intricate semantic reasoning.

As countermeasures to these concerns, we propose implementing a hybrid fact-checking pipeline integrating multistage evidence retrieval and deep claim classifiers.

Specifically, our system combines a pre-trained SBERT model for semantic retrieval with a Cross-Encoder for fine-tuned claim-evidence relevance modelling. To verify claims eventually, we trained a BERT classifier on serialised claims and Top-k evidence. Our system achieved a development set accuracy of 0.4870 and an F1-score of 0.2253, ranking in the Top-15 on the leaderboard and demonstrating the strength of our method.

This report presents our system architecture, methodological decisions, experimental assessments, and results.

## 2 Approach

### 2.1 Baseline Method

In this project, we set a baseline approach which is Term Frequency-Inverse Document Frequency(TF-IDF) with Bidirectional Long Short-Term Memory(BiLSTM). In the retrieval section, TF-IDF got 0.0864 of F score. And in the prediction section, BiLSTM got 0.4351 of accuracy.

### 2.2 Pipeline Overview

Figure 1 illustrates the overall design of our system, which consists of a thorough evidence recovery process and a three-stage retrieval pipeline.

During the data preparation process, raw claim and evidence data are cleaned and preprocessed to ensure consistent formatting and remove noise. An SBERT model is then fine-tuned with Multiple Negatives Ranking Loss(MNRL) for improved semantic representation for retrieval tasks.

The three-stage retrieval pipeline first retrieves the Top-2000 candidates with SBERT-based dense retrieval, filters them down to Top-15 through fast filtering based on precomputed embeddings, and finally reranks the remaining candidates using a fine-tuned Cross-Encoder to select the Top-4 most relevant evidence.

### 2.3 Model Preparation

This section reports the comprehensive realization of each module, from data preprocessing to model fine-tuning, and the multi-stage retrieval pipeline.

#### 2.3.1 Data Preprocessing

Reliable evidence retrieval is grounded on clean and semantically dense textual data. To improve

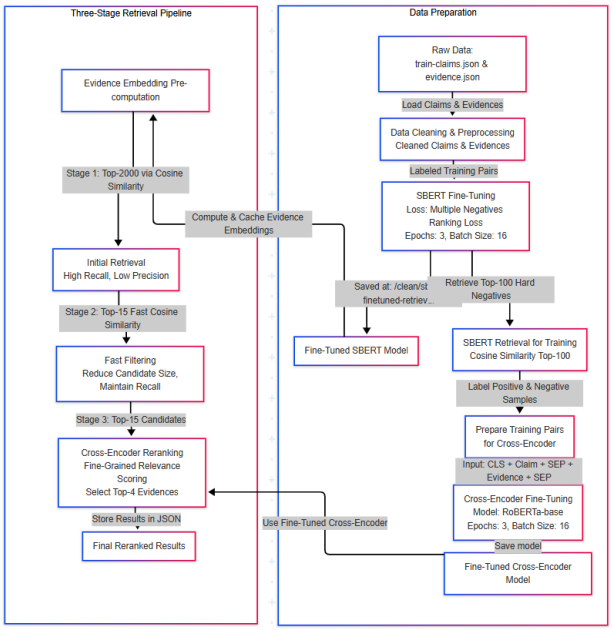


Figure 1: Evidence Retrieval Process

retrieval accuracy, we designed data cleaning to remove irrelevant noise without sacrificing crucial contextual details.

We tried three preprocessing methods: no cleaning, stopwords removal, and retaining stops words after cleaning. As indicated in Table 1, keeping stopwords consistently produced the optimal outcomes.

Removing stopwords is common in most NLP tasks, but experiments here reveal that in climate science data, stopwords play a significant role.They provide contextual fluency and grammatical relationships important for accurate evidence retrieval.

Based on this, we employed a cleaning method that normalizes Unicode (NFKC), removes HTML tags and non-alphanumeric characters, converts text to lower case, and leaves stopwords. This helps maintain the integrity of semantic structures.

| Preprocessing | F-score | Accuracy | Recall |
|---|---|---|---|
| No Cleaning | 0.2204 | 58.44% | 0.2485 |
| Keep Stopwords | 0.2331 | 58.44% | 0.2627 |
| Remove Stopwords | 0.2186 | 55.19% | 0.2464 |

Table 1: Effect of Preprocessing

### 2.3.2 SBERT Fine-Tuning for Retrieval

While fine-tuning SBERT on train claim-evidence pairs, we purposefully experimented with various loss functions to observe how they should be optimized to help improve retrieval performance.

Loss function selection is important to guide SBERT to learn discriminative semantic representations to assist effective evidence retrieval.

We also evaluated multiple standard loss functions rigorously in fine-tuning: Online Contrastive Loss, Triplet Loss, Cosine Similarity Loss, and MNRL. As shown in Table 2, MNRL achieved the best performance, with an F1 score of 0.2403 and recall of 0.2771 which is significant higher than our baseline method(0.0864). This aligns with Reimers and Gurevych's (2019) seminal work on SBERT (Reimers and Gurevych, 2019), which demonstrated that MNRL improves training efficiency by automatically generating negatives within each batch. The MNRL loss is formally defined as follows (Reimers and Gurevych, 2019):

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^{+}))}{\sum_{i=1}^{N} \exp(\text{sim}(\mathbf{q}, \mathbf{d}_i))} \quad (1)$$

Online Contrastive Loss and Triplet Loss suffered from inefficient negative sampling and slower convergence that reduced retrieval performance. Cosine Similarity Loss worked worst, as it failed to include discriminative learning signal critical to guiding effective representation learning.

As a result of these findings, we used MNRL for fine-tuning.

Table 2: Loss Function Comparison

| Loss Function | F-score | Recall |
|---|---|---|
| Online Contrastive Loss | 0.2009 | 0.1280 |
| Triplet Loss | 0.1987 | 0.2240 |
| Cosine Similarity Loss | 0.0795 | 0.0896 |
| Multiple Negatives Ranking | 0.2403 | 0.2771 |

### 2.3.3 Cross-Encoder Reranking

Based on the refined SBERT and training pairs previously established, we utilize a Cross-Encoder to conduct fine-tuned relevance modeling between evidence and claims to enhance evidence screening accuracy. Although SBERT effectively retrieves semantically relevant candidates through self-encoding independently, it is not able to model intricate semantic interactions explicitly. As Reimers and Gurevych noted (Thakur et al., 2020),this bi-encoder limitation weakens relevance ranking, especially when subtle contextual cues determine claim-evidence relationships. Cross-Encoders, in contrast, use complete attention across input pairs to achieve more precise ranking in this regard.

This exploits its two fold advantage of batch-wise negative sampling and linear computational complexity $O(n)$ with the cost of Cross-Encoder architectures being $O(n^2)$. As shown in(Vatolin et al., 2021) points out, Cross-Encoder architecture have $O(n^2)$complexity with big datasets and are therefore unrealistic for real-world evidence retrieval tasks. Therefore, we use the hybrid retrieval process, which combine SBERT with MNRL and Cross encoder to balance speed and accuracy.

We fine-tune a RoBERTa-based SBERT model for top-100 candidate retrieval and reserve the Cross-Encoder for reranking purposes only. This approach allows the model focus on significant tokens and accurately capture minute semantic relations.

#### 2.3.4 Three-Stage Retrieval Pipeline

For a balance between retrieval efficiency and precision across huge evidence collections, we designed a three-stage retrieval pipeline. The approach gradually filters candidate evidence with controlled computational costs, with high recall in early stages and precision in the Top-ranked final list.

**Stage 1 & 2: SBERT Initial Retrieval & SBERT Fast Filtering**

First, we used the fine-tuned SBERT model to obtain top 2000 candidate evidence based on cosine similarity. High recall is preferred here for the sake of not missing any useful evidence even if with lower precision. A larger candidate pool enables better downstream filtering and ranking.

Then, we employ precomputed embeddings to efficiently shortlist the candidates to the top 15 candidate evidence. This highly reduces the computational load of the Cross-Encoder reranking step without damaging important recall.

**Stage 3: Cross-Encoder Reranking**

At the final retrieval pipeline stage, we employ a CrossEncoder fine-tuned to re-rank the top-15 candidate evidence sentences generated by the preceding two stages of retrieval. Unlike SBERT, which fails to capture subtle semantic interaction through separate encoding of claim and evidence, this model jointly consider them to predict subtle semantic relevance—compensating for the inability of SBERT.

Our dataset distribution of evidence (Fig 2) shows most claims are linked with not more than five gold evidence. Due to the characteristics of data distribution, we further tried to verify which one has better effect under different Top-k settings.

Based on the Top-K selection strategy of Yang et al.(Yang et al., 2019), in which both retrieval coverage and accuracy are guaranteed by optimising K, we specifically conducted experiments to analyze optimal end choice size (N). We tested different Top-N sizes from 1 to 5 and verified retrieval performance. As shown in Fig 3, larger N improved recall, yet F1 score gains declined for N=5. Thus, we chose Top-4 to get the trade-off between precision and recall well.

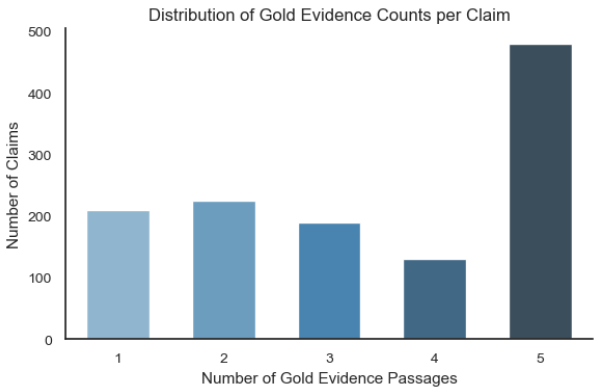


Figure 2: Gold Evidence Counts per Claim

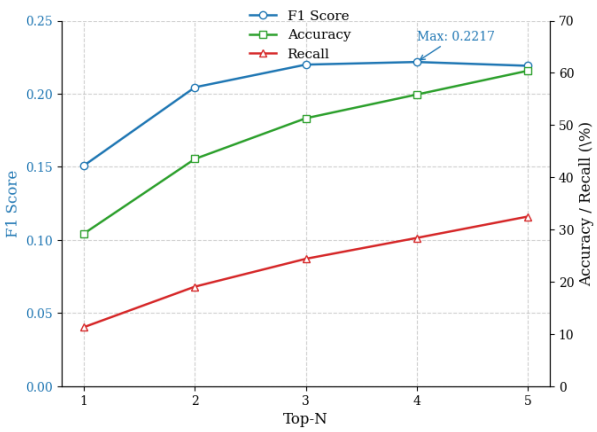


Figure 3: Retrieval Performance wit Top-N Values

### 2.4 Claim Classification with BERT Fine-Tuning

Following the conventional BERT input format (Devlin et al., 2019), we concatenated each claim with its top-ranked evidence using the [SEP] token so that the model can jointly encode claim-evidence interactions for multi-class verification. Based on the processed data, we fine-tuned a BERT model from the Hugging Face Transformers library.

We trained the model for 5 epochs with a learning rate of $3 \times 10^{-5}$ . The final fine-tuned BERT model achieved an accuracy of 0.4870 on the development set.

Otherwise, the model comparisons and detailed performance studies are presented in Section 3.2.

## 3 Experiments

All experiments do not use a fixed random seed, so the randomness in model initialisation and data random sorting may cause small fluctuations in evaluation metrics (such as F1 score) each time they are executed.

### 3.1 Retrieval Improvement Experiment

For F score improvement, we use BM25 after the SBERT Dence Retrieval, then do the CrossEncoder. In this experiment, we improved our F score to 0.242. (In test data, get the 0.19) The probable reason for this result is that SBERT sometimes will pick the overly semantically similar sentence with different keywords. In this scenario, BM25 could help with reranking by the claim keywords to balance out this issue. However, When multiple pieces of evidence are combined, the keyword signal of BM25 is diluted and cannot effectively judge the relevance. In addition, the contextual relationship is ignored. Therefore, the label prediction is also significantly worse. In the final solution we didn't consider this improvement approach.

### 3.2 Model Comparison and Analysis

For the prediction model, we set four experiments, including Multi-Layer Perceptron (MLP) with two hidden layers, Convolutional Neural Network for Text (TextCNN), BiLSTM, and BERT. In the model experiments section, we aim to use different model frameworks to explore their impact on classification performance.

The MLP input is vectorised data by Term Frequency-Inverse Document Frequency (TF-IDF) based on the statistical method used to identify the text's importance across the corpus. As a traditional fully connected network model, MLP can learn the classification logic from keywords, but it can't process the word order in the sentence. In MLP, claim and evidence can only be seen as word combinations, not as complete sentences, leading to the worst accuracy.

On the other hand, word embedding data is used as input for TextCNN. TextCNN can catch the n-gram feature (we set three to five in the experiment), which means the model can get more information than from each individual word. However, it still can't capture long-distance semantic dependencies like BERT.

BiLSTM further improves sequential modelling by processing word embeddings in both forward and backward directions. This allows it to understand sentence-level dependencies better than MLP. However, its ability to model long-range semantic dependencies remains limited compared to the transformer architecture.

For BERT, the input is the raw text with data cleaning. In contrast to other models, the transformer framework uses text is bidirectional, which can capture the sentence's meaning more precisely (Devlin et al., 2019). As a pre-trained model, BERT already knows the standard grammar rules and semantic associations. Based on this, we fine-tune all parameters and add the "self classifier" to initialise the weight and enhance the performance.

As shown in Table 3, the accuracy of 0.4870 for the BERT model is the highest, and the result from the MLP is the lowest. Finally, TextCNN and BiLSTM achieve intermediate accuracy, with BiLSTM performing slightly better due to its sequential modelling capability, but still falling short of BERT's powerful contextual understanding.

Table 3: Performance of Different Models

| Model | Type | Input Feature | Accuracy |
|---|---|---|---|
| MLP | Traditional NN | TF-IDF | 0.3831 |
| CNN | Convolutional NN | Word Embeddings | 0.4416 |
| BiLSTM | Recurrent NN (Baseline) | Word Embeddings | 0.4351 |
| BERT | Transformer | Raw Text | 0.4870 |

### 3.3 Evidence concatenation Strategies for Claim Classification

Based on our experimental results, the multi-evidence concatenation approach achieved the highest claim classification accuracy on the development set (0.4870), outperforming both the logits/probabilities aggregation and majority voting methods. Although majority voting produced similar results on the development set, its accuracy dropped significantly to 0.41 on the test set, indicating likely overfitting to the development data and higher sensitivity to noisy or inconsistent evidence. In contrast, concatenating all selected evidence into a single input allows the model to jointly leverage contextual information from multiple sources, resulting in more stable and robust predictions across

both the development and test sets. Therefore, we ultimately selected the evidence combination method as our final submission.

As shown in Table 4, we systematically explored and compared several evidence aggregation strategies, including majority voting and logits/probabilities fusion, rather than simply following existing routines. This comparative analysis enabled us to thoroughly assess the strengths and limitations of each method. Our findings empirically support that concatenating multiple pieces of evidence as a single BERT input yields more consistent performance, providing new insight into multi-evidence reasoning for fact-checking tasks.

Table 4: Aggregation Strategy Comparison

| Method | Claim Classification Accuracy |
|---|---|
| Multi-Evidence Concatenation | 0.4870 |
| Logits/Probabilities Aggregation | 0.4545 |
| Majority Voting | 0.4740 |

### 3.4 Experimental Creativity and Analysis

Comparing the method with dance similarity, BM25 and cross-encoder, we tried three innovative methods. First, we set a confidence threshold from the cross-encoder. The BM25 and dense full-text were used to get the four evidence that didn't pass the threshold. This test's F score is 0.214. Future improvements will be made to the fallback role, which is more qualified because only BM25 and dense can't improve the retrieval model's performance.

Secondly, we created a multi-score reranking system, which performs a weighted average of the score from the cross-encoder and dense similarity after BM25 and dense initial retrieval. This method got a 0.228 F score. Dynamically adjusting the weights based on the feature of the claim might be a significant direction. The fixed weight can't adapt to diverse samples, which might be the reason for this method's suboptimal performance.

The final experiment uses a simple version of maximal marginal relevance to control the diversity of evidence, which gets the F-score 0.228. The filter standard to get the low similarity evidence might have mistakenly removed the correct options. Adding the weighted redundancy penalty is a possible future improvement direction. Overall, the F scores of these innovative experiments still maintain a certain level, and if continuously improved, there might be a breakthrough.

Table 5: Innovative Method Performance

| Method | F1 Score |
|---|---|
| SBERT Dence Retrieval with BM25 and CrossEncoder rerank | 0.242 |
| CrossEncoder rerank + threshold fallback | 0.214 |
| Multi-Score Re-ranking | 0.228 |
| Light MMR | 0.228 |

## 4 Results and Conclusion

To conclude, while the F1-score on the test set (0.1737) was smaller compared to that on the development set (0.2253). The probably reason is that we did not do the data preprocessing such as removing repetitively evidence id with same content of evidence dataset which leads to the model adaptive lower. On the other hand, accuracy remained similar (0.4935 vs 0.4870), showing prediction model no overfitting.(Table 6) All these results were in line with expectations and show the excellent generalizability of our hybrid fact-checking pipeline. Also comparing to baseline approach improve dramatically and ranked as 15th on leaderboard.

In addition to the main pipeline, our creativity experiments of retrieval and reranking methods currently offer no major gain over our final solution but maintain competitive performance and present promising avenues for future research. One limitation of our study is the lack of a fixed random seed, which may lead to slight fluctuations in the reported metrics. Addressing this issue in future experiments will help improve the reproducibility of the results.

In future work, automatically setting the weight of retrieval evidence for the label prediction model could be a meaningful direction. Because some correct evidence doesn't have valuable information for the prediction section, which might lead to the situation that even if the retrieval result improves, it still can't achieve corresponding improvements in prediction. For improvement, an attention mechanism could be an effective plan.

In this project, we learned to find the possible experimental direction from the previous test. In the exploration, we need to fully understand the interactive influence of model combination and then make the trade-off to get a balanced solution for retrieval and prediction performance.

Table 6: Performance on Dev and Test Sets

| Dataset | Accuracy | F1-Score | Harmonic Mean |
|---|---|---|---|
| Development Set | 0.4870 | 0.2253 | 0.3080 |
| Test Set | 0.4935 | 0.1737 | 0.2569 |

## Team Contributions

- **Yu Han Jen(1508398)**: Responsible for developing, deploying, and experimenting with the main retrieval pipeline of SBERT-based dense retrieval, cross-encoder reranking, and the final best-performing hybrid retrieval model.
- **Chin Hsuan Liang(1528053)**: Responsible for experimenting and testing different methods (MLP, TextCNN, Bert) and new retrieval/reranking methods. Also assisted with comparison experiments as well as ablation studies.
- **Tingfang Zhou (1535837)**: Focused on the baseline (BiLSTM) model code and was primarily based on writing, structuring, and polishing the report, including tables and interpretation of experimental results.

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